

Implementation

Community Detection – Modularity Maximization

- Community structure detection by maximizing modularity
 - dense connections within communities, sparse between communities
- Steps:
 - For every possible merge (or node move between communities), compute the change in modularity if that merge/move is made.
 - Find the merge (or node move) that gives the largest positive and update the community structure accordingly.
 - Recalculate modularity gains for the new community structure, and repeat Step 2 & 3 until no operation can further increase modularity.
 - Among all intermediate partitions during the process, choose the one with the highest modularity as the final community structure.

Community Detection: Louvain

Community Detection using Louvain Method

- Number of Communities: 36
- Modularity Score: 0.828
- Total Internal Edges: 1424
- Avg. Internal Density of Communities: 0.081



Communities	
Community 0	(78 nodes)
Community 31	(59 nodes)
Community 2	(97 nodes)
Community 3	(2 nodes)
Community 5	(24 nodes)
Community 6	(21 nodes)
Community 7	(25 nodes)
Community 9	(24 nodes)
Community 10	(16 nodes)
Community 11	(22 nodes)
Community 1	(30 nodes)
Community 29	(26 nodes)
Community 8	(21 nodes)
Community 18	(22 nodes)
Community 24	(23 nodes)
Community 22	(19 nodes)
Community 32	(19 nodes)
Community 21	(22 nodes)
Community 26	(18 nodes)
Community 27	(20 nodes)
Community 28	(52 nodes)
Community 30	(20 nodes)
Community 14	(45 nodes)
Community 33	(20 nodes)
Community 34	(23 nodes)
Community 4	(25 nodes)
Community 12	(25 nodes)
Community 16	(22 nodes)
Community 15	(25 nodes)
Community 17	(12 nodes)
Community 20	(24 nodes)
Community 13	(8 nodes)
Community 35	(2 nodes)
Community 23	(1 nodes)
Community 25	(1 nodes)
Community 19	(1 nodes)

Community Detection: Leiden

Community Detection using Louvain Method

- Number of Communities: 33
- Modularity Score: 0.789
- Total Internal Edges: 1567
- Avg. Internal Density of Communities: 0.093



Communities	
Community 0	(78 nodes)
Community 31	(59 nodes)
Community 2	(97 nodes)
Community 3	(2 nodes)
Community 5	(24 nodes)
Community 6	(21 nodes)
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Community 19	(1 nodes)

Community Detections: Comparison

	Graph Structural Dismantling	Label Propagation	Modularity Maximization	
Algorithms	Girvan-Newman	LPA	Louvain	Leiden
Number of Comm.	36	72	36	33
Modularity	0.750	0.693	0.828	0.789
Avg. Internal Density	0.073	0.071	0.081	0.093

- Modularity Maximization Method presents a more stable structure and high-quality performance
- **Leiden is adopted in later analysis** as it offers a more balanced overall performance, with a higher modularity, least fragmented structure, and the highest internal cohesion.

Analysis:

Comm. Composition

- The subgraph has no giant clusters, indicating a multi-centric, multi-cluster structure
- Most of the communities range from 20-30 nodes
 - community structure is dispersed with pronounced modularity

	community	num_nodes	num_products	num_customers	pct_products	avg_degree
0	0	97	62	35	0.639175	6.721649
1	1	58	9	49	0.155172	7.120690
2	2	49	23	26	0.469388	2.163265
3	3	48	13	35	0.270833	2.520833
4	4	44	3	41	0.068182	2.204545
5	5	40	13	27	0.325000	4.925000
6	6	37	15	22	0.405405	3.054054
7	7	27	2	25	0.074074	2.296296
8	8	26	20	6	0.769231	2.769231
9	9	26	14	12	0.538462	3.153846
10	10	26	21	5	0.807692	3.000000
11	11	25	19	6	0.760000	3.720000
12	12	25	20	5	0.800000	5.040000
13	13	25	6	19	0.240000	9.280000
14	14	25	3	22	0.120000	3.720000
15	15	25	2	23	0.080000	3.880000
16	16	24	13	11	0.541667	3.333333
17	17	24	7	17	0.291667	4.208333
18	18	24	11	13	0.458333	4.375000
19	19	24	14	10	0.583333	4.958333
20	20	24	6	18	0.250000	2.625000
21	21	24	4	20	0.166667	2.000000
22	22	23	19	4	0.826087	4.260870
23	23	22	15	7	0.681818	5.045455
24	24	22	20	2	0.909091	2.409091
25	25	21	9	12	0.428571	2.142857
26	26	19	17	2	0.894737	5.157895
27	27	18	4	14	0.222222	2.277778
28	28	17	12	5	0.705882	4.117647
29	29	2	1	1	0.500000	1.000000
30	30	1	1	0	1.000000	0.000000
31	31	1	1	0	1.000000	0.000000

Analysis:

Comm. Composition

- **pct(%)_products**: varies from 0.06 to 0.9+
 - “product-heavy vs. customer-heavy” approaches:
 1. **product-heavy**(% > 0.5): A small number of customers connected to a large number of products.
 - Strong category depth & relatively concentrated and possibly repetitive user behavior
 2. **customer-heavy**(% < 0.5): A large number of customers are concentrated around a relatively small number of products
 - Community behavior is determined by “user behavior patterns,” not by product structure
 3. **mixed balanced cluster**(% ~ 0.5)

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Analysis:

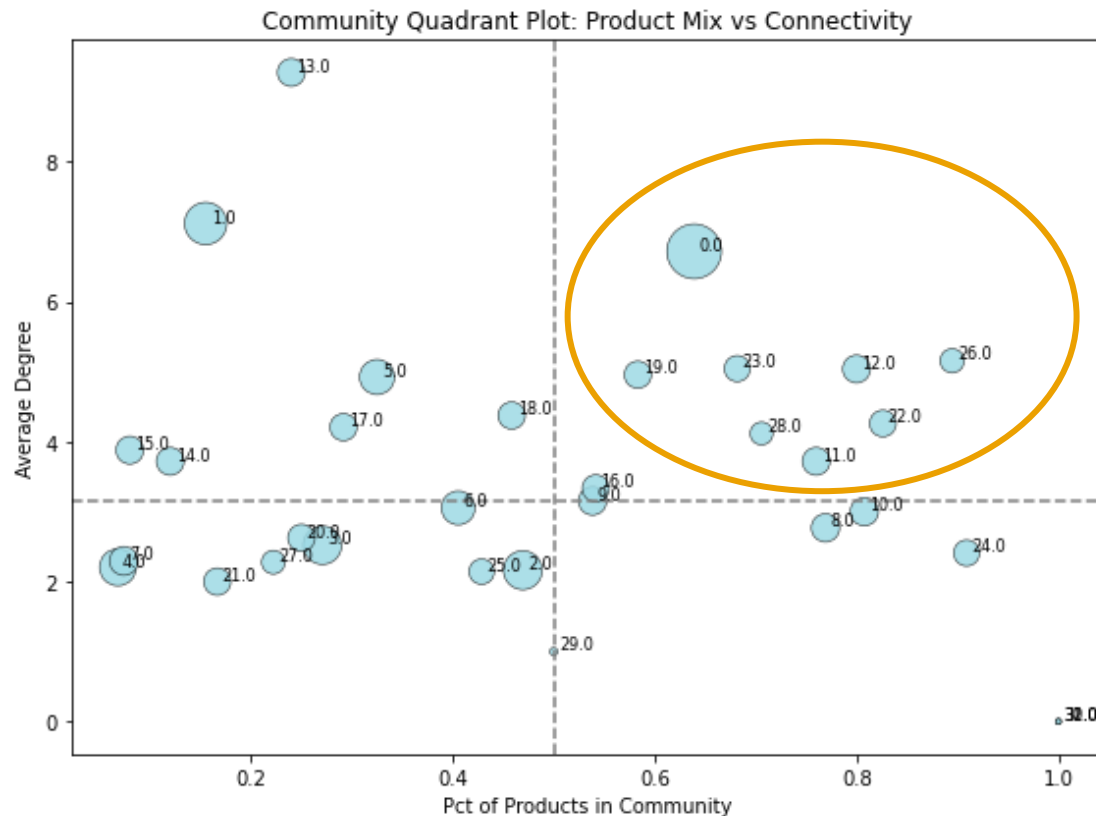
Comm. Composition

- **avg_degree**: extreme variation
 - some communities(>5) are tightly knit while others(<=3) exhibit extremely loose structures
- 1. **high-density**: stable structure with strong cohesion
 - products: reviewed by lots of users
 - customers: reviewing on products frequently/ posting high quality reviews(high numbers of "votes" & "helpful")
- 2. **low-density**: loose structure & diverse community connection --> tends to be "explorative users" or "first-time customers"
 - products: niche products
 - customers: low engagements/ newcomers/ browsers

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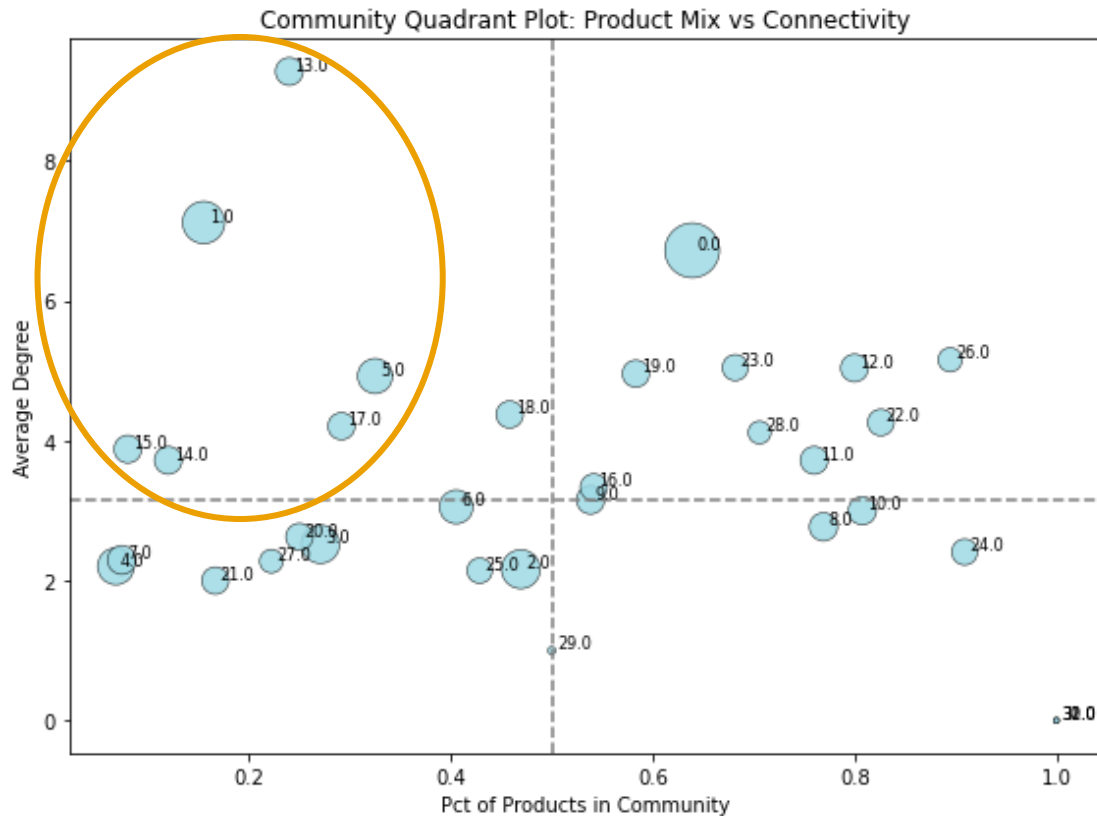
Persona Profiling



- **1. Categorical Core Shoppers / Category Deep Divers:**
 - product-heavy + high density community
 - Example: Community 0 (97 nodes)
- large amount of products & few highly active customers
- category-driven cluster
- **user profiling:**
 - loyal buyers with highly concentrated interests in specific categories
 - long-term customers, not one-time buyers
 - a small set of highly active customers engage with a large set of closely related products
 - a major product category with strong co-purchase or co-review behavior

Analysis:

Persona Profiling

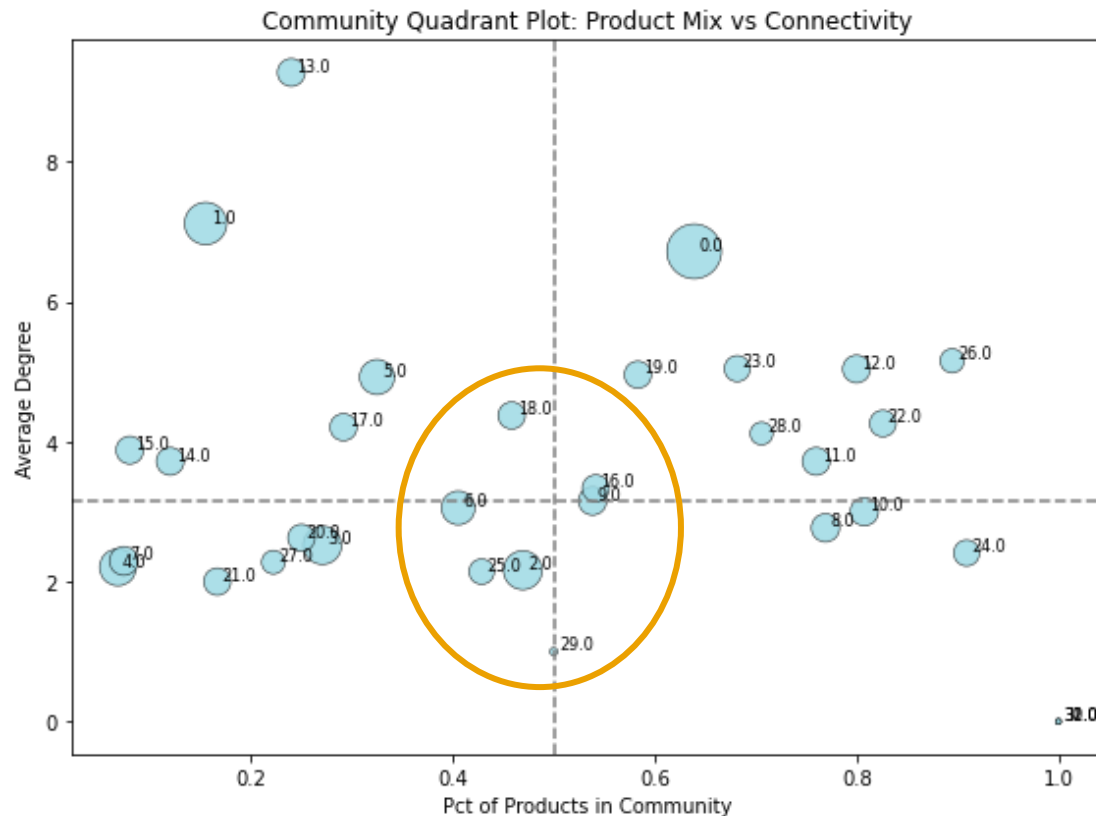


2. Fan Clusters / Star-product Buyers:

- customer-heavy + high avg_degree community
- Example: Community 1 (58 nodes)
- high avg_degree: customers engage intensely, leaving high-quality reviews and browsing on the same groups of products frequently
- strong community-driven behavior, possibly a fandom cluster or viral product segment
- **user profiling:**
 - power influencers
 - high-volume trend followers clustered around a few blockbuster products
 - frequently review on popular contents
 - driven by popularity signals rather than category depth

Analysis:

Persona Profiling



- **3. Casual shoppers:**

- mixed balance + low-engagement & low-inner-bonding community

- Example: Community 2 (49 nodes)

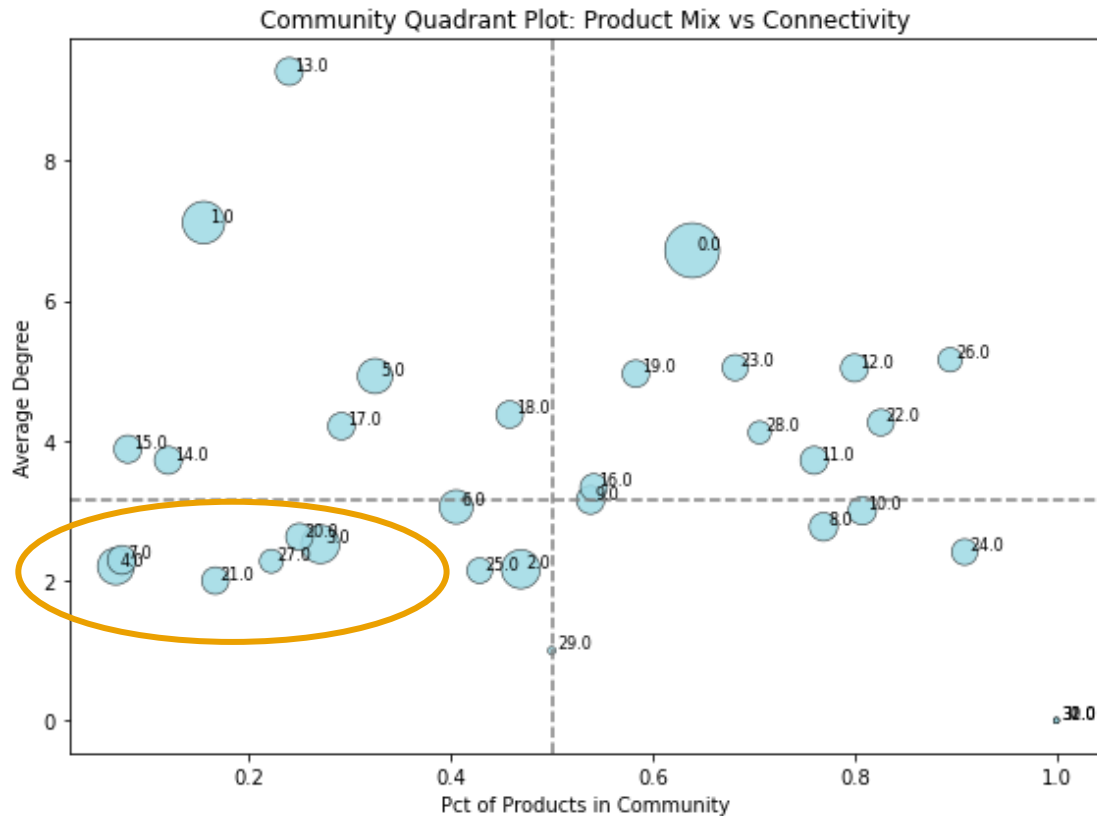
- balanced ratio of products and customers: users interact with a variety of items that are not strongly related to each other.

- **user profiling:**

- non-specialized, low-commitment browsing behavior
- general-purpose shoppers who browse across multiple categories
- customers exploring different types of products

Analysis:

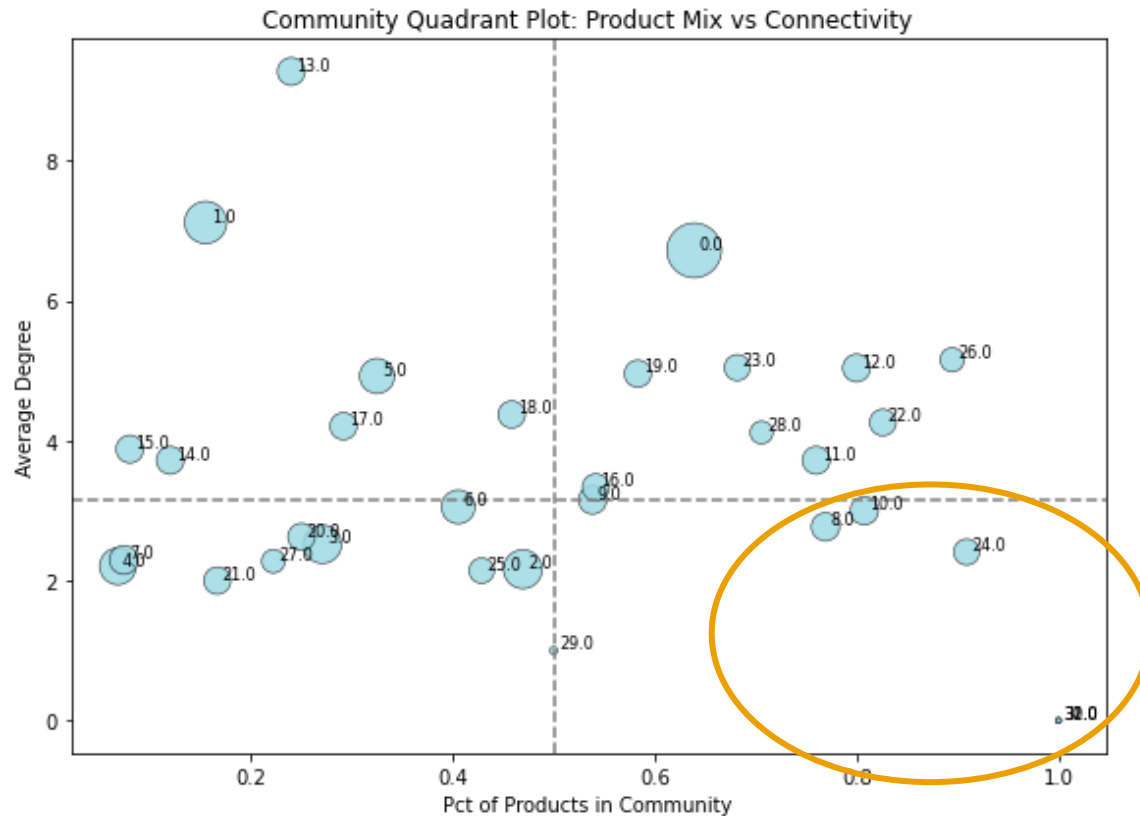
Persona Profiling



- **4. Occasional Buyers / Comparison Shoppers**
 - customer-heavy + low-engagement community
 - Example: Community 3 (48 nodes)
- low interactions & commitments, but sharing overlapping interests
- **user profiling:**
 - new customers entering a specific category for the first time
 - occasional buyers with focused but shallow engagement
 - potentially “one-time demand”
 - customers evaluating a few similar products (substitutes) before making a choice

Analysis:

Persona Profiling



- **5. Browsers / Potential Customers:**

- product-heavy + low-density community
- Example: Community 8 (26 nodes)
- customers browsing various products with low correlations
- **user profiling:**
 - random visitors
 - no specific browsing direction and minor engagement
 - early stage of purchasing initiation

Business Insights: Targeted Strategies

Categorical Core Shoppers	Fan clusters / Star-product Buyers	Casual shoppers	Occasional Buyers	Browsers / Potential Customers
Product-heavy	Customer-heavy	Mix-balanced	Customer-heavy	Product Heavy
High density	High engagement	Low engagement	Low engagement	Low density
In-category deep recommendations	Limited effectiveness of personalized recommendations	Explorable, wide-coverage entry products	not suitable for deep personalization / content-based	
Cross-sell complementary products	Promotion on trending products/boosting viral items	Help users to spark interests on a category, not deepen their understanding	Introduction recommendation of diverse products to capture interests	